

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Internet Programming

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COURSE CODE: CSA43

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Web Development Standards Implementation

Case Study Analysis & Solutions



Scenario

A startup is developing a modern web application. During development, the team encountered several issues related to HTML elements, HTTP methods, XML syntax and web communication protocols.

The following case study identifies the correct technologies, explains the reasons for each selection and demonstrates best practices followed in modern web development.



Question 1

✓ Selected Option : B

Correct HTML Tag for Video Embedding

Answer : `<video src="video.mp4" controls></video>`

The HTML5 **<video>** element is the standard tag used to embed videos in a webpage. The **controls** attribute provides play, pause, fullscreen and volume controls for users.

Question 2

✓ Selected Option : B (POST)

Appropriate HTTP Method for Form Submission

Answer : POST

The **POST** method is used to submit registration form data securely to the server. Unlike GET, POST sends data inside the request body instead of displaying it in the URL. It is commonly used for forms containing usernames, passwords, email addresses and other sensitive information.

Question 3

✓ Selected Option : B (<nav>)

Proper Semantic Navigation Element

Answer : <nav>

The **<nav>** element is a semantic HTML5 tag specifically designed to contain navigation links such as menus, sidebars and page navigation. It improves accessibility, search engine optimization (SEO) and makes the webpage easier to understand.

Question 4

✓ Selected Option : B

Correct XML Syntax

Answer :

```
<note>
  <to>User</to>
  <from>Admin</from>
</note>
```

Errors in Other Options

- Option A - Missing closing </note> tag.
- Option C - Incorrect closing tag (</from> instead of </to>).
- Option D - Missing closing bracket and invalid XML syntax.

Question 5

✓ Selected Option : B (HTTP)

Correct Web Communication Protocol

Answer : HTTP

HTTP (HyperText Transfer Protocol) is the standard protocol used for communication between web browsers and web servers. It enables users to request and receive web pages over the Internet.

FTP is used for file transfer, SMTP is used for email communication and SNMP is used for network management.